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CMPT 477
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Project Two

Image 1: Lena_Y.tif

Original Image	Image w/ Gaussian Noise: 0.2σ	Mean Filter
	 RMSE: 9.082, PSNR: 28.967	 RMSE: 6.473, PSNR: 31.908
Original Image	Image w/ Gaussian Noise: 0.2σ	Smart Filter
	 RMSE: 9.089, PSNR: 29.96	 RMSE: 5.696, PSNR: 33.02

Original Image	Image w/ Gaussian Noise: 0.3σ	Mean Filter
		
	RMSE: 13.615, PSNR: 25.450	RMSE: 7.295, PSNR: 30.87

Original Image	Image w/ Gaussian Noise: 0.3σ	Smart Filter
		
	RMSE: 13.585, PSNR: 25.47	RMSE: 6.847, PSNR: 31.421

Original Image	Image w/ Gaussian Noise: 0.2σ	Better Smart Filter
		
	RMSE: 9.065, PSNR: 28.984	RMSE: 5.544, PSNR: 33.25

Better Smart Filter: $W = [[1, 3, 1], [3, 7, 3], [1, 3, 1]] / 23$

Image 2: Airplane-F16_Y.tif

Original Image	Image w/ Gaussian Noise: 0.2σ	Mean Filter
	 RMSE: 8.864, PSNR: 29.178	 RMSE: 8.220, PSNR: 29.833

Original Image	Image w/ Gaussian Noise: 0.2σ	Smart Filter
	 RMSE: 8.87, PSNR: 29.173	 RMSE: 6.747, PSNR: 31.548

Original Image	Image w/ Gaussian Noise: 0.3σ	Mean Filter
		
	RMSE: 13.289, PSNR: 25.661	RMSE: 8.867, PSNR: 29.175

Original Image	Image w/ Gaussian Noise: 0.3σ	Smart Filter
		
	RMSE: 13.299, PSNR: 25.654	RMSE: 7.692, PSNR: 30.41

Original Image	Image w/ Gaussian Noise: 0.2σ	Better Smart Filter
		
	RMSE: 8.862, PSNR: 29.18	RMSE: 6.315, PSNR: 32.123

Better Smart Filter: $W = [[1, 3, 1], [3, 7, 3], [1, 3, 1]] / 23$

Conclusion:

Mean filtering and smart filtering are two methods that can be used to remove additive noise from an image. However, both methods have the disadvantage of removing details from the image. Smart filters try to mitigate this disadvantage by computing a weighted mean instead of simply an arithmetic mean, however they still suffer from the same issue. Even though the PSNR between the filtered image and the clean image was much better than that of the noisy image, visually the filtered image still is of low quality.